

Lucía Cabrera

luciacabrera.ar



Pick a flower on Earth and
you move the farthest star.
– Paul Dirac

Areas of specialization

• Theoretical & Mathematical
Physics • String Theory &
M-Theory • AI Evaluation &
Benchmarking

Languages

Spanish	C2	mother tongue
English	C2	• • • • •
German	B2	• • • • •
Italian	B2	• • • • •

Programming Languages

L ^A T _E X	• • • • •
haskell	• • • • •
python	• • • • •
lean	• • • • •

Dev Stack

Git	• • • • •
Jupyter Notebooks	• • • • •
OpenAI API	• • • • •



EDUCATION

- 2021 – **Physics Ph.D. Student**
BALSEIRO INSTITUTE · Bariloche, Argentina 📍
Thesis: Particle Physics, Quantum Gravity and Dualities in String Theory.
Supervisor: Prof. Gerardo Aldazabal **Co-supervisor:** Prof. Bobby Acharya
- 2019 **Physics Licentiate Degree**
NATIONAL UNIVERSITY OF ROSARIO · Rosario, Argentina 📍
Thesis: Dual Holographic Description of Yang-Mills Plasmas **Supervisor:**
Prof. Martin Schvellinger (La Plata Institute of Physics - National University
of La Plata). **GPA:** 9.53/10



EXPERIENCE

- 2025 – **AI Evaluator & Benchmarking Researcher**
POINDEXTER LABS · London, United Kingdom 📍
• Designed adversarial math/physics datasets to probe LLM reason-
ing • Built evaluation pipelines (OpenAI API, Bloom); contributed to AI
safety prompting (Forum AI collab) • Analyzed model failure modes &
co-authored methodology paper on prompt sensitivity.
- 2022 – 2025 **Teaching Assistant**
BALSEIRO INSTITUTE · Bariloche, Argentina 📍
• Instructed undergraduate courses in linear algebra, differential equa-
tions, and complex analysis.
- 2016 – 2019 **Teaching Assistant**
NATIONAL UNIVERSITY OF ROSARIO · Rosario, Argentina 📍
• Instructed undergraduate courses in linear/spectral algebra, complex
analysis and distribution theory.



MANUSCRIPTS

- 2026 **Too long; didn't solve · arXiv:2604.07593 [cs.AI]**
L. CABRERA, I. SAXTON-KNIGHT ·
On how prompt length shapes LLM behaviour and discriminativeness.
- In preparation **On Capturing Stringy Features within a Field Theory Description**
G. ALDZABAL, E. ANDRÉS, L. CABRERA, M. MAYO ·
Embedding stringy features into a QFT and its hidden symmetries.
- In preparation **Branches of M-Theory/Heterotic string theory duals**
B. ACHARYA, D. BALDWIN, L. CABRERA, K. NARAIN ·
A classification of Heterotic string theories dual to M-theory on G_2 mani-
folds.

HIGHLIGHTED SCHOLARSHIPS & AWARDS

- 2025 **ZONTA Women in STEM Award – South-American Winner**
ZONTA INTERNATIONAL · 📍
- 2023 **STEP Fellowship**
INTERNATIONAL CENTRE FOR THEORETICAL PHYSICS · Trieste, Italy 📍
Tri-annual international research collaboration fellowship.
- 2020 **Highest GPA in Physics**
PUNTO BIZ MAGAZINE · Santa Fe, Argentina 📍
Highest GPA in a physics degree in the province of Santa Fe, Argentina.
- 2019 **Doctoral Fellowship**
NATIONAL SCIENTIFIC AND TECHNICAL RESEARCH COUNCIL · Ar-
gentina 📍
- 2019 **Summer Research Scholarship**
BALSEIRO INSTITUTE · Bariloche, Argentina 📍
Research internship. **Project:** Communication Servers for LAHN/ANDES.



INVITED TALKS

- 2024 **Invited Talk** · Imperial College London. United
Kingdom.
- 2024 **Contributed Talk** · 5th Young Frontiers Meeting.
Online.
- 2024 **Contributed Talk** · 1st Online Conference for
Women in Theoretical Physics from LATAM. Online.

RESEARCH VISITS

- 2025 **ICTP** - as STEP fellow. Italy.
2024 **ICTP** - as STEP fellow. Italy.
2023 **ICTP** - as STEP fellow. Italy.
2022 **IPhT Saclay**. France.